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Paper 1: Talazoparib Formulation Bridging in Cancer Patients—Challenges and the Critical

Bibliographic Information

- **Title:** Talazoparib Formulation Bridging in Cancer Patients—Challenges and the Critical Role of Model-Informed Drug Development in Approval Despite Failed Bioequivalence
- **Authors:** Wang D, Guo CC, et al.
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Quick Take

This paper demonstrates how model-informed drug development (MIDD) enabled the approval of a new talazoparib soft gelatin capsule (SGC) formulation despite a 37% higher C_{max} that failed bioequivalence. By using a semi-mechanistic PK/PD model to show that AUC (not C_{max}) drives hematologic toxicity, the study was powered only for AUC equivalence, reducing sample size and study duration. The approach provides a template for formulation bridging when C_{max} is highly variable and potentially higher for the new formulation.

Executive Summary

This paper describes the critical role of model-informed drug development (MIDD) in the approval of a new talazoparib soft gelatin capsule (SGC) formulation for commercial use. The initial hard capsule (HC) formulation had been approved, but the SGC formulation required a bioequivalence (BE) and food effect (FE) study. Challenges included a long half-life (~90 h), high variability in C_{max} (intra-patient CV 40%), and the need for steady-state assessment in cancer patients. Using a semi-mechanistic PK/PD model (Friberg model) for hematologic toxicity (anemia, neutropenia, thrombocytopenia), the authors demonstrated that increased C_{max} (up to 52%) does not affect safety when AUC is unchanged. This supported a BE study design powered only for AUC equivalence (intra-patient CV 17%), reducing the required sample size from 160 to 56 patients and shortening study duration from 140 to 56 days. The study confirmed AUC bioequivalence (90% CI 99.00–111.70) but C_{max} was 37% higher (90% CI 125.05–149.27). Safety data showed no clinically significant differences. The totality of evidence (AUC equivalence, lack of C_{max} impact on safety, and observed safety) supported FDA approval of the SGC formulation. This work exemplifies how MIDD can overcome regulatory and operational challenges in formulation bridging.

Scientific Context & Motivation

The problem addressed is the need to bridge a new soft gelatin capsule (SGC) formulation to the existing hard capsule (HC) formulation for talazoparib, a PARP inhibitor with a long half-life and high pharmacokinetic variability. Traditional bioequivalence studies require demonstration of both AUC and C_{max} equivalence, but C_{max} is highly variable (intra-patient CV 40%) and the SGC dissolution data suggested potentially higher C_{max} . Additionally, the study must be conducted at steady state in cancer patients, making a single-dose crossover unethical. The knowledge gap was how to design a feasible study when C_{max} is likely to fail equivalence. The paper fills this gap by using MIDD to establish that AUC (not C_{max}) drives the main safety concerns (myelosuppression), allowing the study to be powered only for AUC equivalence. This paradigm challenges the traditional requirement for C_{max} equivalence in post-approval formulation changes, demonstrating that exposure-response relationships can justify alternative endpoints.

✂ Methodological Snapshot

The study used a population PK model (previously published) and a semi-mechanistic PK/PD model (Friberg model) for myelosuppression (hemoglobin, neutrophils, platelets) to evaluate the impact of C_{max} on safety. The BE study was a 2-period crossover design with sample size based on AUC equivalence only (intra-patient CV 17%). Treatment duration was optimized using PK simulations: 28 days for Period 1, 21 days for subsequent periods. PK parameters were estimated using noncompartmental analysis. BE was assessed using a mixed-effects model on log-transformed data with 90% CIs.

Detailed Methodological Analysis

Modeling Approach

Population PK model (from prior work) and semi-mechanistic PK/PD model (Friberg model) for hematologic endpoints. The PK/PD model used a linear treatment effect on progenitor cell proliferation. Simulations were performed by manually changing model parameters (e.g., K_a , CL) to create scenarios with different C_{max}/AUC combinations.

Data Sources

PK/PD model: data from 466 patients in Phase 1-3 studies (PRP-001, 673-201, 673-301) with talazoparib monotherapy 0.025 – 1.1 mg QD. BE study: 73 patients with advanced solid tumors, 53 PK-evaluable. PK sampling: pre-dose and 0.5 – 24 h post-dose on last day of each period.

Estimation Methods

PK parameters calculated using noncompartmental analysis (Open NCA). BE assessed using mixed-effects model on log-transformed data (SAS or similar). PK/PD model parameters estimated using NONMEM (not specified in detail).

Model Evaluation

PK/PD model was evaluated using visual predictive checks (not shown) and simulations. The BE study used 90% CIs and prespecified acceptance criteria (80 – 125%).

Covariate Analysis

Not explicitly performed; the PK/PD model used population typical values. The simulation considered patients with typical and low clearance (5th percentile) to assess impact on steady-state achievement.

Statistical Rigor Assessment

The sample size calculation was based on intra-patient variability from a prior DDI study (40% for C_{max} , 17% for AUC). The 90% power to meet 80 – 125% acceptance criteria is standard. The use of a mixed-effects model on log-transformed data is appropriate. The simulation-based approach to determine treatment duration is rigorous, considering both typical and low clearance patients. The study was open-label, which is acceptable for BE studies. The safety comparison is limited by small sample size but is consistent with the known safety profile.

Key Findings

Primary findings: (1) The SGC formulation met prespecified bioequivalence criteria for AUC (90% CI 99.00–111.70) but C_{max} was 37% higher (90% CI 125.05–149.27). (2) Semi-mechanistic PK/PD modeling showed that increasing C_{max} by up to 52% (while keeping AUC constant) does not significantly affect hemoglobin, neutrophil, or platelet levels, whereas a 25% increase in AUC with a minor C_{max} increase caused substantial myelosuppression. (3) Steady-state C_{trough} was comparable between formulations (GMR 95.33%, 90% CI 89.02–102.09). (4) Food effect on SGC was consistent with HC: high-fat meal decreased C_{max} by 42% but did not affect AUC . (5) Safety profiles were similar between formulations in the first period, with no new signals. These findings support that AUC is the driver for hematologic toxicity and that the higher C_{max} of the SGC formulation does not pose additional safety risks.

Clinical & Regulatory Implications

The SGC formulation can be taken with or without food (similar to HC). No dose adjustment is needed. The higher C_{max} of SGC does not require additional safety monitoring. The study supports the commercial use of the SGC formulation. For other drugs, the approach demonstrates that when C_{max} is not a driver for safety, BE studies can be powered for AUC only, reducing patient burden and development costs. The food effect results support that SGC can be taken without regard to food, consistent with the HC label. Regulatory approval was granted based on the totality of evidence, including the MIDD analysis.

Strengths & Limitations

Strengths

- Use of a well-established semi-mechanistic PK/PD model (Friberg model) to justify the AUC -only endpoint, reducing regulatory risk.
- Population PK simulations to optimize treatment duration (14 days for subsequent periods) and minimize dropout.
- Prospective regulatory alignment through pre-submission meetings and an MIDD meeting with FDA.
- Clear demonstration that the totality of data (AUC equivalence, lack of C_{max} impact on safety, observed safety) supports bridging.
- Significant reduction in sample size (56 vs. 160) and study duration (56 vs. 140 days) compared to traditional designs.
- Realistic eligibility criteria and PK evaluability criteria based on modeling to account for dose interruptions and missed doses.

Limitations (Acknowledged by Authors)

- Small sample size for safety comparison (n=38 for HC, n=35 for SGC) limits the ability to detect rare adverse events.
- Crossover design means only Period 1 adverse events can be fully attributed to each formulation.
- The PK/PD model was developed from a different patient population (Phase 1-3 studies) and may not fully capture the BE study population.
- The simulation scenarios for C_{max} increase were based on manual parameter changes (e.g., increasing K_a) and may not reflect all possible mechanisms.
- The study was open-label, which could introduce bias in safety assessments.

Limitations (Expert Review)

- The PK/PD model used a linear treatment effect on progenitor cell proliferation; a more complex model (e.g., with feedback loops) might have shown different results.
- The 37% C_{max} increase observed in the BE study exceeded the 52% simulated maximum; the model's robustness for this magnitude of increase is not fully validated.
- The study did not assess efficacy endpoints; the assumption that AUC drives efficacy is based on prior knowledge but not directly tested in this bridging study.
- The PK/PD model was not externally validated for the SGC formulation; the simulations assumed similar absorption kinetics.
- The food effect assessment was limited to a single high-fat meal; the impact of other meal types was not evaluated.

Generalizability

The framework is applicable to other drugs where the exposure-response relationship for safety and efficacy is well characterized. It is particularly relevant for drugs with high PK variability, long half-lives, and patient populations where traditional BE studies are infeasible. The approach

requires prior knowledge of the exposure metric driving toxicity and efficacy, which may not be available for all drugs.

Figures & Tables

- **Figure 1:** Comparison of dissolution profiles of talazoparib SGC and HC formulations in 0.01M HCl with paddles at 75 rpm.
 - *Significance:* Shows that SGC dissolves faster than HC, suggesting potential for higher C_{max} and justifying the need for BE study.
- **Figure 2:** Study schema showing the 2-period crossover design for BE (Periods 1 and 2) and a fixed-sequence period for FE (Period 3).
 - *Significance:* Illustrates the optimized study design with 28-day first period and 21-day subsequent periods, as well as the FE assessment.
- **Figure 3:** Simulated PK and PD profiles (hemoglobin, neutrophils, platelets) for four hypothetical subjects with different C_{max} and AUC scenarios.
 - *Significance:* Demonstrates that increased C_{max} (up to 52%) does not affect hematologic toxicity when AUC is unchanged, while increased AUC (25%) causes substantial myelosuppression. This supports the AUC -only endpoint for BE.
- **Table 1:** Summary of drug properties and their impact on study design challenges (long half-life, high variability, clastogenic nature).
 - *Significance:* Provides rationale for why a traditional BE study is infeasible and why MIDD is needed.
- **Table 2:** Simulation results showing AUC ratios on Day 42 (Period 2, Day 14) vs. Day 56 (Period 2, Day 28) for different bioavailability changes.
 - *Significance:* Demonstrates that 14-day treatment in subsequent periods is sufficient to characterize bioavailability differences, supporting the shortened treatment duration.
- **Table 3:** Comparison of sample size and study duration for different BE study designs (AUC -only vs. $AUC + C_{max}$, various crossover designs).
 - *Significance:* Shows that the AUC -only design requires 56 patients vs. 160 for the full replicate design, highlighting the efficiency gain from MIDD.
- **Table 4:** Bioequivalence and food effect results: adjusted geometric mean ratios and 90% CIs for AUC , C_{max} , and C_{trough} .
 - *Significance:* Key results table showing AUC met BE criteria (90% CI 99.00 – 111.70) but C_{max} failed (90% CI 125.05 – 149.27), and food effect on C_{max} is consistent with HC.
- **Table S1:** Baseline demographics and patient characteristics of the safety analysis population.
 - *Significance:* Provides context for the study population and helps assess generalizability.
- **Table S2:** Descriptive statistics of PK parameters by treatment group.
 - *Significance:* Shows intra-patient variability (15% for AUC , 22.7% for C_{max}) and total variability (32 – 40% CV).
- **Table S3:** Comparison of all-grade TEAEs between HC and SGC formulations in Period 1.
 - *Significance:* Supports the safety comparison between formulations, showing no clinically significant differences.

- **Table S4:** Comparison of Grade ≥ 3 TEAEs between HC and SGC formulations in Period 1.
 - *Significance:* Further supports the safety comparison, showing similar rates of severe adverse events.
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Supplementary Materials

Supplementary materials include Appendix S1 (detailed methods for PK evaluability criteria, dosing considerations, and study conduct), Tables S1-S4 (baseline demographics, PK descriptive statistics, and safety comparisons), and Figure S1-S2 (study design schemas and steady-state achievement plots). These provide additional detail on the study design and results.

Future Directions

Future studies could validate the PK/PD model prospectively with the SGC formulation in larger patient populations. The approach could be extended to other PARP inhibitors or drugs with similar pharmacokinetic challenges. Long-term safety data (e.g., beyond 28 days) would strengthen the conclusion that C_{max} does not affect toxicity. The methodology could be applied to other formulation changes (e.g., from immediate-release to modified-release) where C_{max} may differ.

Expert Commentary

This is a textbook example of MIDD in regulatory decision-making. The use of a semi-mechanistic myelosuppression model to justify an AUC -only bioequivalence endpoint is innovative and aligns with the FDA's 'totality of evidence' approach. The study design optimization using population PK simulations to shorten treatment periods is rigorous and practical. The authors' proactive engagement with regulators through pre-submission meetings and an MIDD meeting is commendable. One caution: the PK/PD model was developed from a different dataset and may not fully capture the SGC formulation's absorption characteristics; however, the consistency of the food effect results with the HC formulation provides some reassurance. This work sets a precedent for future formulation bridging studies in oncology.

Bottom Line

For pharmacometricians, this paper provides a compelling example of how MIDD can overcome bioequivalence challenges by leveraging exposure-response relationships. The key lesson is that when C_{max} is highly variable and likely to fail equivalence, a well-validated PK/PD model can justify focusing on AUC only, provided that AUC is the driver for safety and efficacy. This approach reduces study burden and enables regulatory approval despite failed C_{max} . The framework should be considered for other drugs with similar properties, especially in oncology where patient enrollment is challenging.

☒ Figures

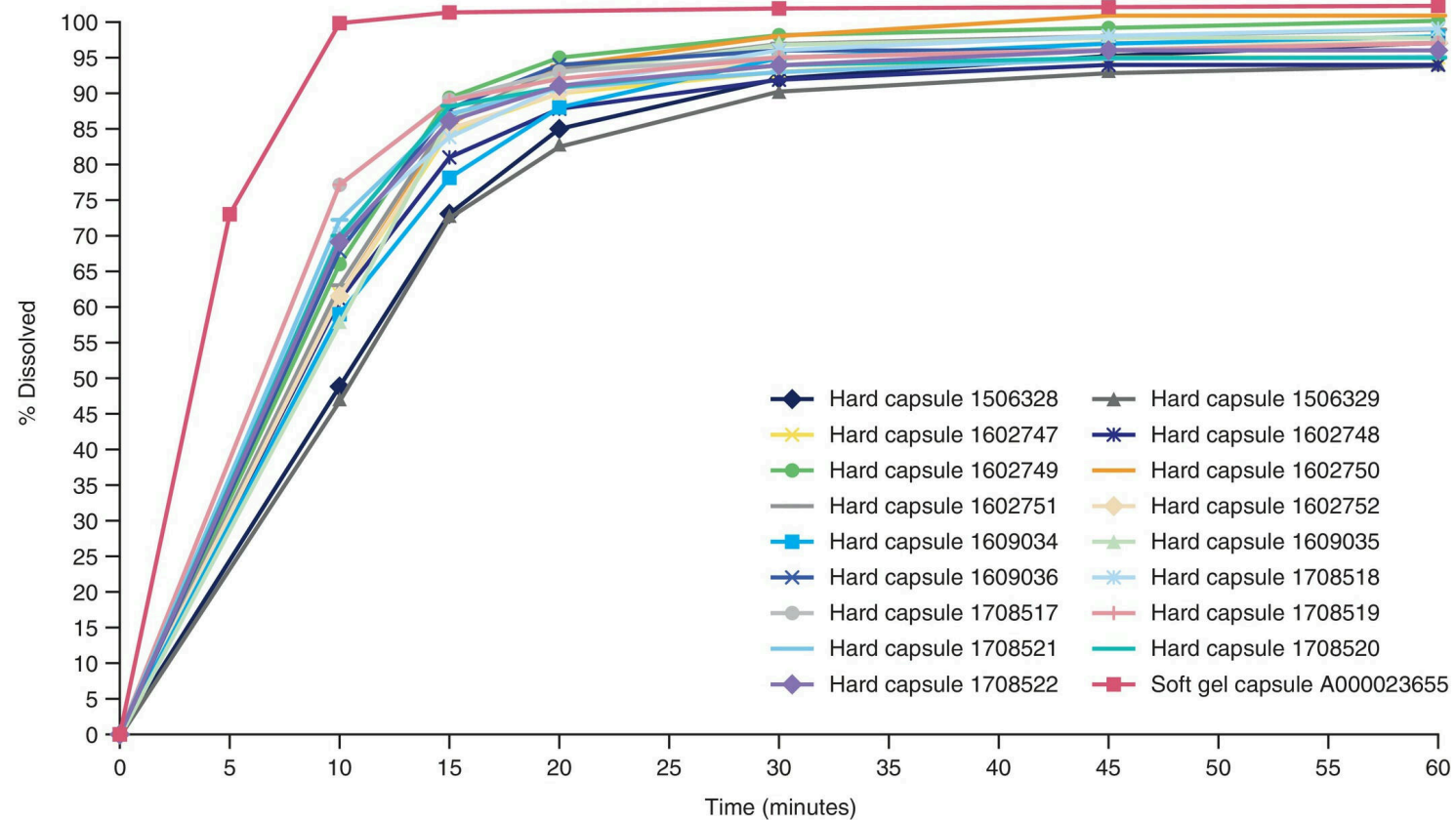


Figure 1: Comparison of the dissolution profile of the talazoparib soft gel capsule with hard shell capsule batches in 0.01M HCl with paddles at 75 rpm. aHCl, hydrochloric

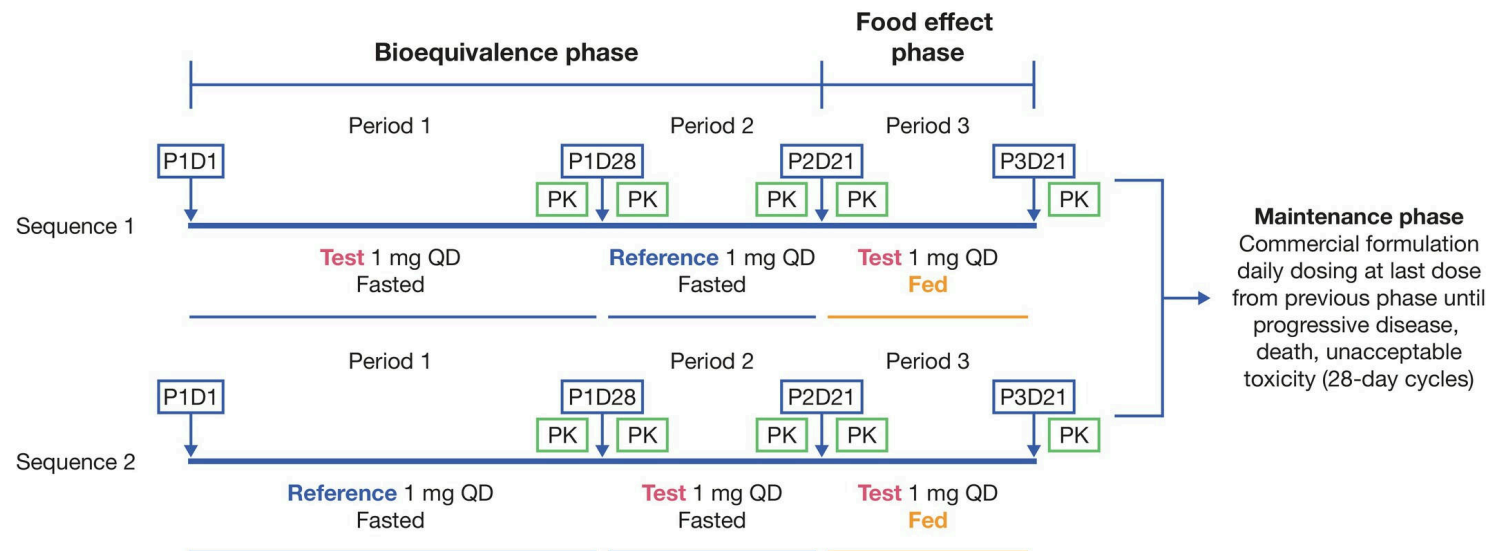


Figure 2: Study schema. D1, Day 1; D21, Day 21; D28, Day 28; P1, period 1; P2, period 2; P3, period 3; PK, pharmacokinetic; QD, once daily.

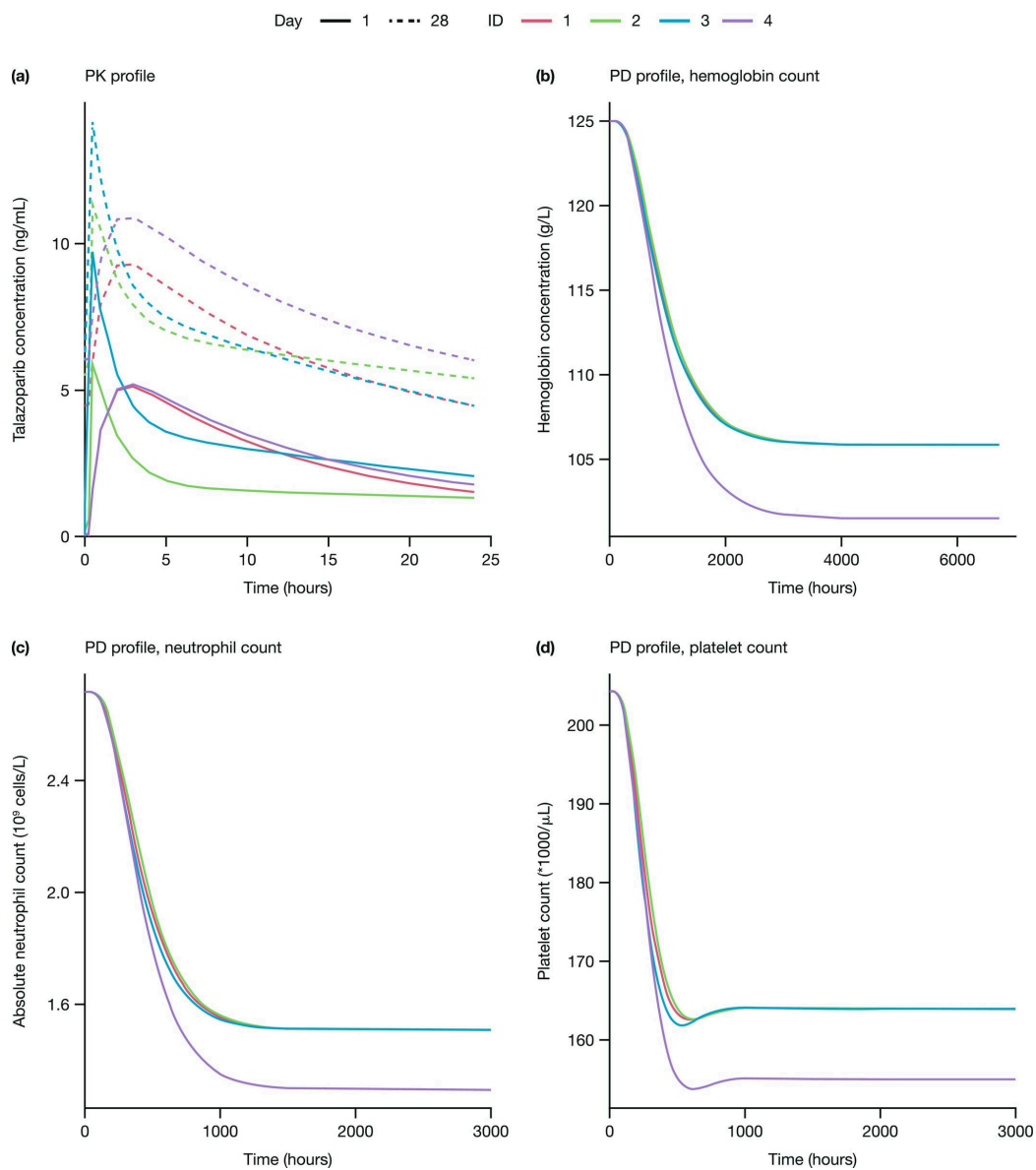


Figure 3: Simulated PK and PD profiles after treatment with talazoparib 1 mg once daily (a) PK profile, (b) PD profile, hemoglobin count, (c) PD profile, neutrophil count,